Exercise on Equation Solutions with minimum constraint

- How many solutions does the equation

$$x_1 + x_2 + x_3 = 5$$

have, where x_1, x_2, and x_3 are nonnegative integers and $x_3 \geq 3$?

- Equivalent problem: Suppose that a cookie shop has 3 different kinds of cookies. In how many different ways can 5 cookies be chosen, if I must choose at least 3 cookies of the third kind?
- **Solution**
- I rewrite the equation as follows:
- $x_1 + x_2 + (x_3 - 3) = 5 - 3 \Rightarrow x_1 + x_2 + x = 2.$
- The number of solutions for the new equation is
- $C(3 + 2 - 1, 2) = C(4, 2) = \frac{4!}{2!2!} = 6$
- This is the number of solutions when $x \geq 0 \Rightarrow x_3 - 3 \geq 0 \Rightarrow x_3 \geq 3.$